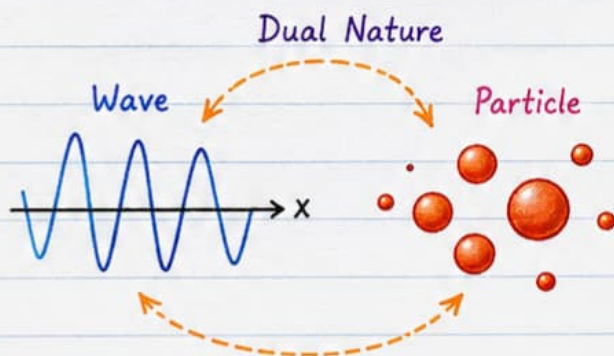


DUAL NATURE OF RADIATION AND MATTER

“Radiation behaves as a wave, matter behaves as a particle, but both reveal their dual nature in the quantum world.”

1 INTRODUCTION

- In classical physics, **light is a wave** and **matter is a particle**.
- But in 20th century, it was discovered that both radiation and matter show **dual behaviour**.
- This concept is the foundation of **quantum mechanics**.

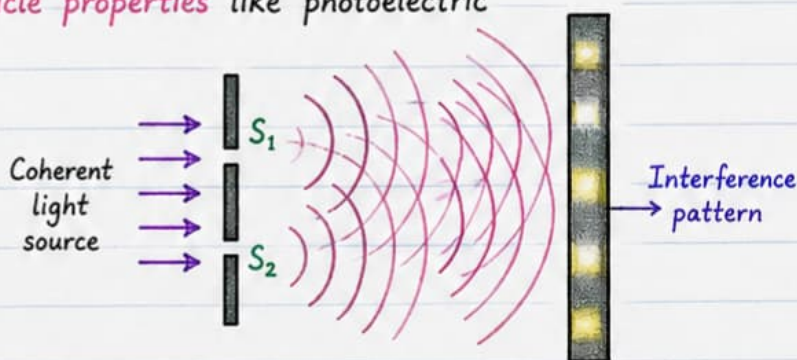


2 DUAL NATURE OF RADIATION (LIGHT)

- Light exhibits **wave properties** like interference, diffraction, polarization and also **particle properties** like photoelectric effect, Compton effect.

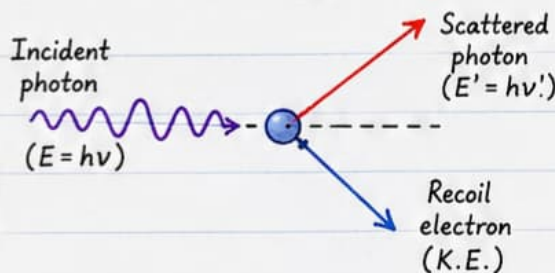
(a) Wave Nature of Light

- Interference
- Diffraction
- Polarization



(b) Particle Nature of Light

- Photoelectric Effect → Light ejects electrons from metal surface.
- Compton Effect → X-rays collide with electrons and change their wavelength.



★ KEY FORMULA

- Energy of photon : $E = h\nu$
- Momentum of photon : $p = \frac{h}{\lambda}$
- Planck's Constant : $h = 6.626 \times 10^{-34} \text{ Js}$

where, ν = frequency, λ = wavelength



DID YOU KNOW?

Einstein explained the photoelectric effect by assuming light is made of packets called **photons**. This idea won him the **Nobel Prize** in **1921**.

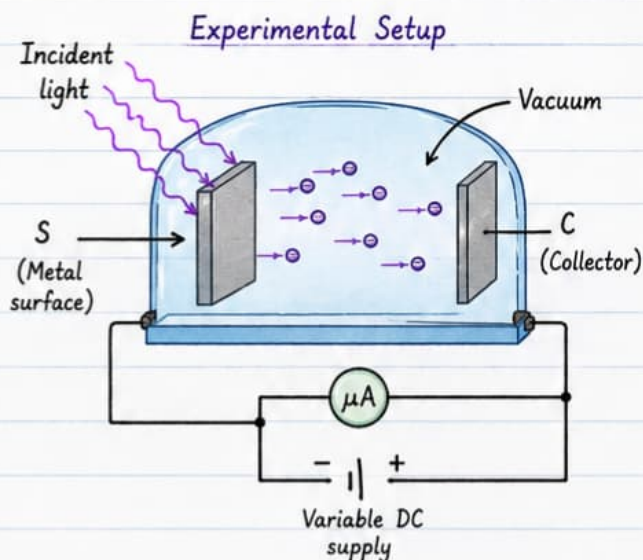


3 PHOTOELECTRIC EFFECT

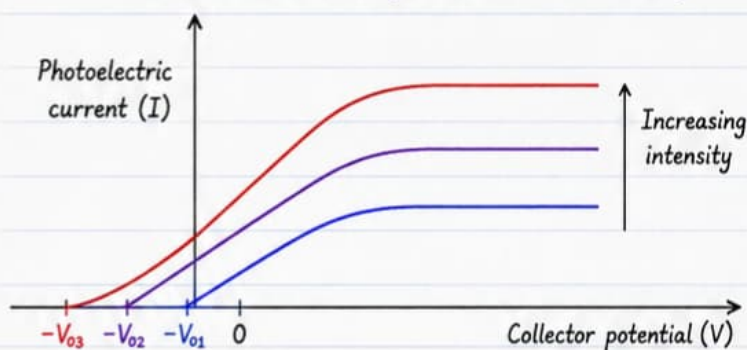
- When light of **suitable frequency** strikes a metal surface, electrons are emitted from the surface. These electrons are called **photoelectrons**.
- This effect can be explained **only if light behaves as a stream of particles** called **photons**.

PHOTOELECTRIC OBSERVATIONS

- For a given frequency (ν) above a certain limit, photoelectric current **increases with intensity**.
- For a given intensity, photoelectric current is **independent of frequency** (above threshold frequency).
- There exists a minimum frequency (**threshold frequency ν_0**) below which no electrons are emitted.
- The maximum kinetic energy of photoelectrons **increases with frequency** and is independent of intensity.



V-I Characteristics (for different intensities)



Stopping potential (V_0) is the negative potential that just stops the most energetic electrons.

EINSTEIN'S PHOTOELECTRIC EQUATION



This equation shows both wave (ν) and particle ($h\nu$) nature of light.

Energy of incident photon = Work function of metal + Maximum kinetic energy of photoelectrons

$$h\nu = \phi + K_{\max}$$

where, h = Planck's constant
 ν = frequency of light
 ϕ = work function of metal
 $K_{\max}$ = maximum kinetic energy

IMPORTANT RELATIONS

- $K_{\max} = eV_0$ (e = charge of electron)
- $eV_0 = h\nu - \phi \rightarrow V_0 = \frac{h}{e} \nu - \frac{\phi}{e}$
- Threshold frequency, $\nu_0 = \frac{\phi}{h}$
- Wavelength form, $K_{\max} = h \left(\frac{c}{\lambda} \right) - \phi$ (c = speed of light)

WORK FUNCTION (ϕ)

Minimum energy required to remove an electron from the surface of the metal.

★ It depends on the nature of the metal.

★ Unit : joule (J) | Also expressed in eV

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$



EXAMPLES OF WORK FUNCTION (APPROX.)

Metal	ϕ (eV)	Metal	ϕ (eV)
Sodium (Na)	2.28	Iron (Fe)	4.50
Potassium (K)	2.30	Copper (Cu)	4.70
Zinc (Zn)	4.31	Platinum (Pt)	5.60



Different metals require **different minimum energy** to emit electrons.



Remember : Light is both wave and particle.
 Use the right model for the right phenomenon!

4 COMPTON EFFECT

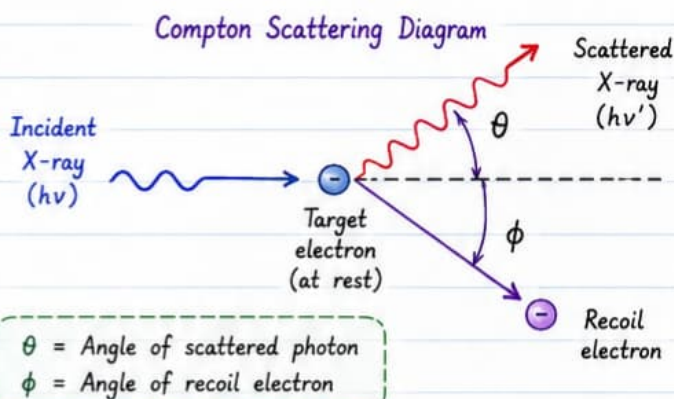
- In 1923, A.H. Compton observed that when X-rays are scattered by free or loosely bound electrons, the scattered X-rays have **longer wavelength** than the incident X-rays.
- This phenomenon is called **Compton effect**.
- It provides strong evidence for the **particle nature** (photons) of light.

Compton Shift Formula

$$\lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta)$$

where,

- λ = wavelength of incident X-ray
- λ' = wavelength of scattered X-ray
- h = Planck's constant (6.626×10^{-34} Js)
- m_e = mass of electron (9.11×10^{-31} kg)
- c = speed of light (3.0×10^8 m s⁻¹)
- θ = angle of scattering



Compton Effect Observations

- Scattered wavelength (λ') is always greater than incident wavelength (λ).
- Wavelength shift ($\lambda' - \lambda$) depends **only on the angle θ** , not on the intensity of X-rays.
- Maximum shift occurs at $\theta = 180^\circ$ (back scattering).**
- No shift occurs at $\theta = 0^\circ$ (forward scattering).**



Compton effect **cannot** be explained by wave theory, but is easily explained using **photon concept**.

5 DE BROGLIE HYPOTHESIS (MATTER WAVES)

- In 1924, Louis de Broglie proposed that not only radiation, but **matter** also exhibits wave nature.
- According to de Broglie, every **moving particle** is associated with a wavelength.
- This wavelength is called **de Broglie wavelength**.

Example Calculation

Particle	Momentum (kg m s ⁻¹)	Wavelength (m)
Electron ($v = 2 \times 10^6$ m/s)	1.82×10^{-24}	3.64×10^{-10}
Alpha particle	1.32×10^{-21}	5.02×10^{-13}
Cricket ball	0.5	1.32×10^{-33}

de Broglie Wavelength Formula

$$\lambda = \frac{h}{p}$$

where,

- λ = de Broglie wavelength
- h = Planck's constant
- p = momentum of the particle ($p = mv$)

Important Points

- Heavier the particle or higher the velocity $\Rightarrow$ shorter wavelength.
- For macroscopic objects, wavelength is extremely small (too small to observe).
- For microscopic particles (e^- , p , n , atoms), wavelength can be measurable.

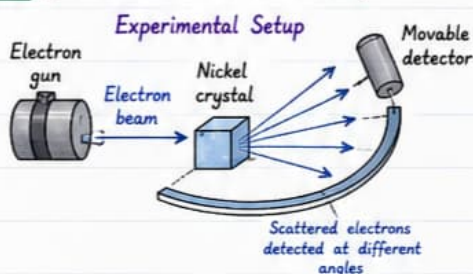


Significance

- Unified the **wave** and **particle** nature of light and matter.
- Laid the foundation of **quantum mechanics**.
- Explains electron diffraction, Davisson - Germer experiment, etc.

6 DAVISSON-GERMER EXPERIMENT (EVIDENCE OF MATTER WAVES)

- Davisson and Germer (1927) proved that **electrons** exhibit wave nature.
- When a beam of electrons is incident on a nickel crystal, **diffraction pattern** of electrons is obtained.
- This pattern is similar to X-ray diffraction, confirming **de Broglie hypothesis**.



Result

- Intensity of scattered electrons varies with angle.
- Maxima and minima obtained $\Rightarrow$ **Diffraction pattern**.
- Confirms that **electrons behave as waves**.



Dual nature shows that reality at microscopic level is governed by **quantum laws**.
Waves and particles are two sides of the same coin!



7 EVIDENCE OF DUAL NATURE

• Radiation shows Particle Nature

- ★ Photoelectric effect
- ★ Compton effect
- ★ Blackbody radiation (Planck's theory)



These phenomena can be explained only if light consists of particles (photons) with energy $E = h\nu$

• Matter shows Wave Nature

- ★ de Broglie hypothesis
- ★ Davisson-Germer experiment
- ★ Electron diffraction



These phenomena can be explained only if matter (particles) has wave-like properties with wavelength $\lambda = \frac{h}{p}$

Think! ?

Why both are needed?

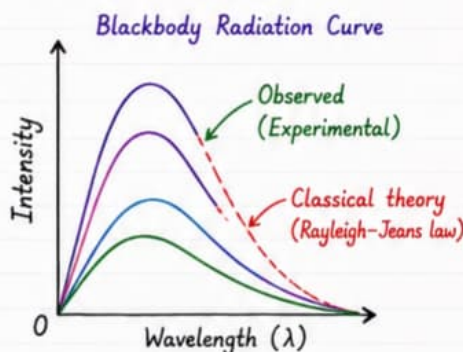
Because in **microscopic world**, neither purely wave theory nor purely particle theory can explain all observations.



Nature is neither only a wave nor only a particle, but **BOTH!**

8 BLACKBODY RADIATION AND PLANCK'S QUANTUM THEORY

- Classical theory failed to explain the observed spectrum of blackbody radiation (Rayleigh-Jeans law led to "ultraviolet catastrophe").
- In 1900, Max Planck proposed that energy is emitted or absorbed in **DISCRETE PACKETS** called **QUANTA**.



• Planck's Postulate

Energy of oscillator can have only certain discrete values.

$$E_n = nh\nu$$

where, $n = 0, 1, 2, 3, \dots$

ν = frequency

h = Planck's constant



Planck's quantum theory introduced the concept of **quantization of energy**, which laid the foundation of quantum mechanics.

9 SUMMARY OF IMPORTANT FORMULAE

Phenomenon / Concept	Formula	Description
Energy of photon	$E = h\nu$	Energy of a photon of frequency ν
Momentum of photon	$p = \frac{h}{\lambda} = \frac{E}{c}$	Momentum of photon
Photoelectric equation	$h\nu = \phi + K_{\max}$	Einstein's photoelectric equation
Threshold frequency	$\nu_0 = \frac{\phi}{h}$	Minimum frequency to eject electrons
Stopping potential	$eV_0 = K_{\max}$	V_0 is stopping potential
Compton shift	$\lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta)$	Change in wavelength after scattering
de Broglie wavelength	$\lambda = \frac{h}{p} = \frac{h}{mv}$	Wavelength associated with a moving particle

KEY POINTS

- ★ Light behaves as wave in interference, diffraction and polarization.
- ★ Light behaves as particle in photoelectric and Compton effects.
- ★ Matter behaves as particle in everyday life.
- ★ Matter behaves as wave in de Broglie and electron diffraction.

Dual nature is the essence of **QUANTUM WORLD**.



QUICK REVISION

Light $\rightarrow$ Wave
Interference
Diffraction
Polarization

Light $\rightarrow$ Particle
Photoelectric effect
Compton effect

Matter $\rightarrow$ Particle
Mass, inertia,
rest mass,
Newton's laws

Matter $\rightarrow$ Wave
de Broglie equation,
Electron diffraction

Result $\rightarrow$
Both are complementary
and necessary.

Understand the dual nature and you will understand the universe deeper!

Keep Learning,
Keep Exploring!

Remember: Quantum world is strange, but it is **REAL** — and it rules the microscopic universe!

